The filling step of Theorem 7: $\mathrm{ext}(\alpha)$ is an equivalence

the mathematics of `seg/model/GExtEquiv2.seg`

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This note states, as a mathematical argument, what the file `GExtEquiv2.seg` proves. It follows the cleaned definition, in which a cofibration is a single decidable sieve rather than a list of generators; the file is being brought in line with it. Each paragraph corresponds to a group of declarations of the file; their names are given in grey. A declaration marked `opaque` in the file is a lemma here: after it is stated, only its statement is used.

Setting

The base is a small category $\mathcal{C}$ with an object $[1]$ and, for every object X, two maps $e_0, e_1 : X \to [1]$, natural in X (the constant maps at the end points). We write $\mathbb{I} = y[1]$ for the presheaf represented by $[1]$, so that $\mathbb{I}(W) = \mathrm{Hom}(W, [1])$. We do not assume that $\mathcal{C}$ has products: posets and Δ are both instances. For each object X there is a set $\Phi(X)$ of decidable sieves on X, stable under precomposition. A map $f : V \to X$ *lands in* $Q \in \Phi(X)$ if $f \in Q$.

- G is a presheaf (the context) and A is a type over G. We write $A(X, \rho)$ for its fibre at $\rho \in G(X)$, and E_A for the total presheaf of the pairs (ρ, a).
- A *cofibration* ψ over G is a decidable sieve $\psi_\rho \in \Phi(X)$ for each $\rho \in G(X)$, natural in ρ: $\psi_{\rho f} = f^*\psi_\rho$. The sets $\Phi(X)$ are closed under finite unions. The point ρ is *full*, written $\mathrm{full}(X, \rho)$, if $\mathrm{id}_X \in \psi_\rho$. By naturality, a map $f : W \to X$ is full (meaning ρf is full) exactly when $f \in \psi_\rho$. Being full is a proposition.
- $G|\psi$ is the context of the full points. B is a type over $G|\psi$, and $\alpha : B \to A|\psi$ is a map over $G|\psi$.

Lines. A *line in A over $\rho \in G(X)$* is a family $\ell(W, f, j) \in E_A(W)$, for $f : W \to X$ and $j \in \mathbb{I}(W)$, which lies over ρf and is natural:

$$\ell(W, f, j) \cdot g = \ell(V, fg, jg) \qquad (g : V \to W).$$

It is thus a map $yX \times \mathbb{I} \to E_A$ over ρ, which is how the family path object is defined without products in $\mathcal{C}$. Its ends are $\ell(e_0) = \ell(X, \mathrm{id}, e_0)$ and $\ell(e_1) = \ell(X, \mathrm{id}, e_1)$. This is the family path object (`GFam.famPath`). The *constant line* at ρ is the line of the base G given by $(W, f, j) \mapsto \rho f$ (`constF`).

Fibres and equivalences. Let $m : B \to A$ be a map over a context D. For $w \in E_A(X)$, an element of the *fibre* $\mathrm{Fib}_m(X, w)$ is a pair (b, ℓ). Here $b \in B$ lies over the point of w, and ℓ is a line in A over that point with $\ell(e_0) = m(b)$ and $\ell(e_1) = w$ (`GEquiv.HFib`: `fibB`, `fibL`, `fibL-0`, `fibL-1`).

An *equivalence structure* on m (`EquivOp`) extends every partial element of a fibre to a total one. The extension agrees with the sides (`kside`), is uniform (`kunif`), and is the value of any side that contains the identity (`kfull`).

The data. The file works with the following data.

- An equivalence structure e_α on α (`eal`).
- A filling structure k_A for A at e_1 (`kA : FibAt e1`). It extends a partial line in A, over a line of G, whose end at e_1 is given totally, to a total line with the same end. Uniformity and agreement with the sides are as above.

$\mathrm{Ext}(\alpha)$**.** Let $\rho \in G(X)$. If ρ is full, $\mathrm{Ext}(\alpha)(X, \rho) = B(X, \rho)$. Otherwise it is the set of pairs (a, b) (`GExt`, `Pairs`) where:

- $a \in A(X, \rho)$;
- b assigns to each full $f : W \to X$ an element $b_f \in B(W, \rho f)$;
- b is natural: $b_f \cdot \chi = b_{f\chi}$;
- $\alpha(b_f) = a \cdot f$.

The map glue sends such a pair to $\mathrm{Ext}(\alpha)(X, \rho)$ in both cases. If ρ is full, it takes b_{id}. The map $\mathrm{ext} : \mathrm{Ext}(\alpha) \to A$ satisfies $\mathrm{ext}(\mathrm{glue}(a, b)) = a$ (`glue-ext`). Because glue handles both cases, the argument below never needs to ask whether ρ is full.

The problem

Fix X and $w = (\rho, a) \in E_A(X)$. Let p be a partial element of $\mathrm{Fib}_{\mathrm{ext}}(X, w)$. It has sides $Q'_0, \ldots, Q'_{n-1} \in \Phi(X)$ and values $p_l(V, f) \in \mathrm{Fib}_{\mathrm{ext}}(V, wf)$ for f landing in Q'_l. These values are natural in f and agree on overlaps.

We construct an element $v_0 \in \mathrm{Fib}_{\mathrm{ext}}(X, w)$ (`v0`). In `GExtEquiv3` it is shown that v_0 extends p and is uniform; this proves that $\mathrm{ext}(\alpha)$ is an equivalence (half of Theorem 7). The idea is a single filling in A, over the constant line at ρ. One family of sides comes from p. The other comes from extending, with e_α, what p says at the full maps.

The argument

Construction 1 (θ, `th`)**.** Let $f : W \to X$ be full. Restrict p along f. At a full point, a fibre element of ext is a fibre element of α: on the B side, $\mathrm{Ext}(\alpha)$ is B there (`toFA`, `Tr`). This gives a partial element of $\mathrm{Fib}_\alpha(W, wf)$. Let

$$\theta_f \in \mathrm{Fib}_\alpha(W, wf)$$

be its extension by e_α (`kE`).

Lemma 2 (th-nat, th-full, th-cong)**.**

1. θ is natural: $\theta_f \cdot \chi = \theta_{f\chi}$ for $\chi : W' \to W$.
2. If f lands in Q'_l, then θ_f is $p_l(W, f)$ read as a fibre element of α.
3. θ_f does not depend on the proof that f is full.

Proof. 1. Restricting the translated partial element along χ gives the translation of the restricted one. The claim then follows from the uniformity of e_α (kE-act).

2. Along f, the side Q'_l contains the identity of W (lands-idr). An extension is the value of a side that contains the identity (kE-full). Restricting along the identity changes nothing (pv-hom).
3. Being full is a proposition.

□

Construction 3 (the base point, x0)**.** The line of G is the constant line at ρ. Its prescribed total end at e_1 is w (x0-E). Its restriction along f ends at wf (x0-act).

Construction 4 (the sides)**.**

(S1) For each side Q'_l of p, take the line of $p_l(V, f)$ (S1). It lies over ρf, ends at wf at e_1 by fibL-1, and is natural in f.

(S2) On the sieve ψ_ρ, take the image in A of the line of θ_f (S2). It is defined because every $f \in \psi_\rho$ is full. The line lies over ρf and ends at wf. It is natural in f by Lemma 2(1).

Lemma 5 (compatibility: c11, c22, c12)**.** The sides (S1) and (S2) agree wherever two of them are both defined.

Proof. Two elements over the same point, whose lines take the same values, are equal (tsEq). So it is enough to compare lines.

- Two sides (S1): this is the compatibility of p.
- The side (S2) with itself: this is Lemma 2(3).
- One side of each kind: f lands in Q'_l and in ψ_ρ. By Lemma 2(2), θ_f is $p_l(V, f)$, and the line of a translated fibre element is the line it came from (toFA-L).

□

The sides are joined into one partial line J with $n + 1$ sides (GCofib.Join).

Construction 6 (the filler, fill)**.** Let $F = k_A(x_0, J)$. This is a line in A over the constant line at ρ. Its values are $F(W, f, j) \in A(W, \rho f)$ (Fv, Fn, Fpt).

Lemma 7 (F1, F-S1, F-S2, F0-act)**.**

1. $F(e_1) = w$.
2. On Q'_l, F is the line of p_l.
3. On ψ_ρ, F is the line of θ.

4. $F(e_0) \cdot f = F(W, f, e_0)$.

Proof. Items 1 to 3 hold because k_A keeps the given end and agrees with the sides (`Fside`), and because the joined partial line restricts to the sides of p and to the side ψ_ρ (`J.atL`, `J.atR`). Item 4 is the naturality of F. □

Construction 8 (the pair, `u`)**.** Let $a_0 = F(e_0) \in A(X, \rho)$ (`a0`). For full f, let b_f be the B component of θ_f (`bE`). Then $u = (a_0, b)$ is a pair (`mkPairs`):

- b is natural by Lemma 2(1) (`pnat0`);
- $\alpha(b_f) = a_0 \cdot f$ (`pal0`) by the chain

$$\alpha(b_f) \overset{\text{fibre}}{=} \text{line of } \theta_f \text{ at } e_0 \overset{\text{L.7(3)}}{=} F(W, f, e_0) \overset{\text{L.7(4)}}{=} a_0 \cdot f,$$

  where Lemma 7(3) applies because the full map f lies in ψ_ρ.

Construction 9 (the fibre element, `v0`)**.** Let $v_0 = (\text{glue}(u), F)$. It lies in $\text{Fib}_{\text{ext}}(X, w)$. Indeed F is a line over ρ, with

$$F(e_0) = a_0 = \text{ext}(\text{glue}(u)) \quad (\texttt{u-ext},\ \texttt{F0}) \qquad \text{and} \qquad F(e_1) = w \quad (\text{Lemma 7(1)}).$$

Remark. The box proof (`ExtEquiv2`, `ExtKit`) filled a box over the cylinder $X \times [1]$, an object of $\mathcal{C}$ for posets. Here a single partial *line* over the constant line is filled. So the argument uses only the presheaf $yX \times \mathbb{I}$, never a product in $\mathcal{C}$, and it applies both to posets and to Δ. No $\forall$ along the interval is needed, because the line of G is constant: the side (S2) is ψ_ρ itself. That quantifier enters only in the composition part of Theorem 7 (step 5).

Correspondence

mathematics	`GExtEquiv2.seg`
θ_f; Lemma 2	`th`; `th-nat`, `th-full`, `th-cong`
base point (constant line, end w)	`cL`, `x0`, `x0-E`, `x0-act`
sides (S1), (S2)	`S1[l].SO`, `S2.SO` (`GMapLine.LineSide`)
compatibility	`tsEq`, `c11`, `c22`, `c12`; `J`
filler F; Lemma 7	`fill`, `Fv`; `F1`, `Fside`, `F-S1`, `F-S2`, `F0-act`
the pair u	`a0`, `bE`, `pal0`, `pnat0`, `u`
v_0	`u-ext`, `gb`, `F0`, `v0`